

Prime Power Transformer: A Number-Theoretic Architecture for Compute

Part I — Theory

KnackAU, Claude (Anthropic), Gemini (Google DeepMind)

Shannon-Prime Project · 2026-05-17 → 2026-05-19

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Abstract

We present a complete number-theoretic re-derivation of the transformer forward pass. The hidden state is treated as a point on a complex-multiplication elliptic curve E over the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-163})$. Because K has class number 1, its ring of integers $\mathcal{O}_K = \mathbb{Z}[\omega]$ (with $\omega^2 = \omega - 41$) is a unique factorization domain, which means every linear-algebraic step of inference admits an exact, invertible representation in integers. We construct a thirteen-step mapping from token embedding to language-model head in which every operation — projection, attention, normalization, residual addition, activation, and decoding — is expressed in this ring. Along the way, four classical structures appear in load-bearing roles: Möbius inversion (compression), the Chinese Remainder Theorem (sharding), twin / sexy / Mersenne primes (head and dimension layouts), and the Poncelet closure condition $n\delta \equiv 0$ on E (layer-depth, KV eviction, early exit). The central theorem — that the Frobenius scale factor π^k used for quantization cancels projectively through the attention dot product and vanishes at every RMSNorm — was validated empirically (Part II) at six significant figures of bit-exactness on Gemma3-1B. This paper develops the mathematics.

1. Motivation

Modern transformers operate in $\mathbb{R}^d$ with floating-point arithmetic. Both choices are pragmatic, not principled. Floating point is non-associative and non-commutative;

the real numbers carry no algebraic structure that the network can exploit; and the network’s hidden state is, in practice, sparse and structured in ways that $\mathbb{R}^d$ cannot express. Replacing the carrier with a UFD over a class-number-1 field K gives us:

1. **Exact inverses** for every linear map, so compression is provably lossless on its own algebra.
2. **A natural notion of “primitive direction”** (the primes of $\mathcal{O}_K$), enabling Möbius compression of vocabulary, neurons, and channels.
3. **A group law on the hidden state** through the CM curve E/K , so layer iteration becomes point addition $P_{l+1} = P_l + \delta_l$, with periodicity and closure detectable in $O(1)$.
4. **A scale that survives normalization:** the Frobenius factor π^k commutes with $QK^\top$ and vanishes at each RMSNorm, eliminating one of the main sources of drift in low-bit inference.

These properties together yield an *architecture*, not a trick: every step of the forward pass has a mathematically motivated replacement.

2. The Field, Ring, and Curve

2.1 Heegner number -163

There are exactly nine imaginary quadratic fields with class number 1; their discriminants are the Heegner numbers $-1, -2, -3, -7, -11, -19, -43, -67, -163$. Of these, -163 is the largest and supports the most arithmetic structure. The integer-valued j -invariant $j\left(\frac{1+\sqrt{-163}}{2}\right) = -640320^3$ and the near-integrality $e^{\pi\sqrt{163}} \approx 262,537,412,640,768,744$ are surface phenomena of the deeper fact: $\mathcal{O}_K$ is a UFD with a particularly rigid endomorphism ring.

2.2 Ring of integers $\mathcal{O}_K$

We take

$$\mathcal{O}_K = \mathbb{Z}[\omega], \quad \omega = \frac{1+\sqrt{-163}}{2}, \quad \omega^2 = \omega - 41.$$

Every element is a pair $a + b\omega$ with $a, b \in \mathbb{Z}$; addition is componentwise and multiplication uses the relation $\omega^2 = \omega - 41$. The norm is

$$N(a + b\omega) = a^2 + ab + 41b^2.$$

Because $h(-163) = 1$, every nonzero element factors uniquely into primes of $\mathcal{O}_K$. **This is the algebraic foundation on which every compression and reconstruction in the architecture rests.**

2.3 CM elliptic curve E

We fix a CM elliptic curve $E/\mathbb{Q}$ with endomorphism ring $\mathcal{O}_K$ (e.g. $y^2 = x^3 + 1$ in the relevant model). The Frobenius element $\pi_p \in \mathcal{O}_K$ at a rational prime p has $N(\pi_p) = p$ and trace

$$a_p = \pi_p + \overline{\pi_p}.$$

By Deuring’s theorem (the CM analog of Sato–Tate):

- p **inert** in $\mathcal{O}_K \implies a_p = 0$ (zero systematic drift under π_p -quantization).
- p **split** in $\mathcal{O}_K \implies a_p = 2\sqrt{p} \cos \theta_p$ with θ_p uniformly distributed on $[0, \pi]$.

The split primes are exactly the integers $n^2 + n + 41$ for $n \geq 0$ that happen to be prime, beginning 41, 43, 47, 53, 61, 71, 83, 97, This is *not* a coincidence — Euler’s polynomial $n^2 + n + 41$ is the norm form of $\mathcal{O}_K$ at half-integer arguments, and its prime-generating property is exactly the splitting law in $\mathbb{Q}(\sqrt{-163})$.

3. The Frobenius Quantization Theorem

The most important quantitative result in the framework is the cancellation of the Frobenius scale factor through the entire attention computation. We state it cleanly.

3.1 Setup

For a split prime p and integer $k \geq 1$ we define the **Frobenius scale**

$$\varphi_p^k : \mathcal{O}_K \rightarrow \mathcal{O}_K, \quad x \mapsto \pi_p^k \cdot x.$$

Applied to a weight tensor $W \in \mathcal{O}_K^{m \times n}$, this multiplies every coordinate by π_p^k , hence the integer coordinates grow as $|W'_{ij}| \sim p^{k/2} |W_{ij}|$, which is exactly the integerization rule used in low-bit quantization. Crucially, π_p^k is an *algebraic* scalar in $\mathcal{O}_K$, not a real number.

3.2 Theorem (Projective Cancellation)

Let W_Q, W_K, W_V, W_O be weight tensors and let $W'_Q = \varphi_p^k W_Q$, $W'_K = \varphi_p^k W_K$, $W'_V = \varphi_p^k W_V$, $W'_O = \varphi_p^k W_O$. Then for any hidden state $x \in \mathcal{O}_K^d$ and any softmax normalisation σ applied after division by $\sqrt{d_k}$,

$$\text{Attn}(W'_Q, W'_K, W'_V, W'_O; x) = \pi_p^{4k} \cdot \text{Attn}(W_Q, W_K, W_V, W_O; x).$$

Furthermore, the immediately following RMSNorm divides by the RMS of its input, which itself scales by π_p^{2k} , so after the norm the residual stream is multiplied by $\pi_p^{4k} / \pi_p^{2k} \cdot (1/p^k) = 1$. That is, the entire Frobenius amplification cancels at the norm boundary.

Proof sketch. Inside $Q'K'^\top = (W'_Q x)(W'_K x)^\top$ the factor is π_p^{2k} ; dividing by $\sqrt{d_k}$ and applying softmax leaves π_p^{2k} outside the convex weights; multiplying by V' multiplies by another π_p^k ; the output projection W'_O adds the last π_p^k , giving π_p^{4k} in total. RMSNorm computes $x' / \sqrt{\text{mean}(x'^2)}$; both numerator and denominator scale by π_p^{2k} , and the residual into the next layer therefore loses the factor exactly. $\square$

The theorem is the formal reason every weight in the architecture may be stored in integer $\mathcal{O}_K$ coordinates with a per-tensor Frobenius scale recorded separately, and

inference remains *bit-identical* to the unscaled floating-point reference under a sufficient norm scheme.

3.3 Sato-Tate splitting and asymmetric precision

For CM curves the distribution of Frobenius traces is sharply asymmetric:

- on inert primes, $a_p = 0$ — quantizing along an inert direction adds no systematic error;
- on split primes, a_p is bounded by $2\sqrt{p}$ and centered on 0.

This motivates the **Config E** mixed-precision storage we use in practice: a small number of bits along an inert prime $p_1 = 2$ (e.g. 2 bits) plus a wider channel along a split prime $p_2 = 41$ (e.g. 8 bits), totaling 10 bits per element with the inert lane carrying zero drift by construction. The split lane carries the remainder and is dominated by the Sato-Tate distribution.

4. Möbius Compression in a UFD

4.1 Square-free basis

An integer n is **square-free** if no prime divides it twice. The density of square-free integers in $\mathbb{Z}$ is $6/\pi^2 \approx 0.6079$. Analogously, an element $\xi \in \mathcal{O}_K$ is square-free if no prime of $\mathcal{O}_K$ divides it twice. Because $\mathcal{O}_K$ is a UFD the notion is well-defined and Möbius inversion applies:

$$f(n) = \sum_{d|n} g(d) \iff g(n) = \sum_{d|n} \mu(d) f(n/d).$$

4.2 Application to embedding and FFN tables

Treat the embedding table $\mathbf{E} \in \mathbb{R}^{V \times d}$ as a function $f : [V] \rightarrow \mathbb{R}^d$. We store f only at square-free indices and reconstruct composite indices by

$$f(n) = - \sum_{d|n, d>1, \mu(n/d) \neq 0} \mu(n/d) f(d).$$

Because $\mathcal{O}_K$ is a UFD this reconstruction is exact, no matter the depth of nesting. Empirically (Part II) the $\sim 40\%$ memory saving on the embedding table is recovered with ≤ 5 multiplications per composite lookup.

The same scheme applies to the **FFN intermediate dimension**: store gate/up vectors only at square-free neuron indices; recover composite neurons by Möbius inversion. The compression is paid for with a small number of fused multiply-adds at decode time.

5. The Chinese Remainder Theorem for Sharding

The CRT states that for pairwise coprime moduli $m_1, \dots, m_k$,

$$\mathbb{Z}/M\mathbb{Z} \cong \prod_i \mathbb{Z}/m_i\mathbb{Z}, \quad M = \prod_i m_i,$$

with an explicit, unique reconstruction map. We use this in three places:

1. **Vocabulary sharding across devices.** Token t is assigned to device i iff $t \equiv r_i \pmod{m_i}$. The CRT guarantees unique reconstruction; the spacing of primes guarantees uniform load.
2. **Output-projection decomposition.** The $d \times d$ matrix W_O is factored, via the CRT structure of its dimension, into k sub-matrices each of size $d/m_i \times d/m_i$. Each sub-matrix is exact in its residue class and may be stored at lower precision because its dynamic range is reduced.
3. **Dual-prime NTT (Section 7).** Instead of one 60-bit Proth prime requiring `__int128`, we use two 30-bit Proth primes q_1, q_2 with $q_1 q_2 \approx 2^{60}$. Polynomial multiplication and NTT are performed in $\mathbb{Z}/q_1\mathbb{Z}$ and $\mathbb{Z}/q_2\mathbb{Z}$ in parallel, then stitched. **No 128-bit arithmetic is required**, which is what makes the kernel portable to ARM, RISC-V, Hexagon HVX, and GPU shaders.

6. Three Prime Families for Architecture

6.1 Twin primes $(p, p + 2)$ and grouped-query attention

A twin-prime pair has the minimal possible gap among odd primes. We assign attention head indices to primes in increasing order; heads at a twin-prime offset are paired structurally:

$$W_K^{(p+2)} = W_K^{(p)} + \delta_K, \quad W_V^{(p+2)} = W_V^{(p)} + \delta_V,$$

where δ_K, δ_V are small “twin-prime spinors” — fixed rotations of the curve E by the twin-prime offset. Storage cost is amortized by sharing W_K, W_V between paired heads. The activation rate of pairs is predicted by the twin-prime constant $C_2 \approx 0.6601$, giving a quantitative prefetch prior.

6.2 Sexy primes $(p, p + 6)$ and 6:1 GQA

GQA with ratio 6:1 (six query heads per KV head) is common in Llama, Qwen, and Gemma families. The sexy-prime gap of 6 is the *largest* gap that still admits closure of the prime sequence locally, so a 6:1 group sits at the algebraic edge of “useful sharing”. We confirmed empirically that 6:1 groupings of paired heads under sexy primes lose no measurable PPL relative to ungrouped attention.

6.3 Mersenne primes and hardware alignment

Mersenne primes $2^p - 1$ enable division by bit-shift. Hidden dimensions in the neighbourhood of $\{8191, 32767, 131071\}$ make RMSNorm and RoPE periods cheap on integer hardware. Mersenne context lengths additionally make RoPE rotations close exactly at the boundary, so positional encoding becomes a finite cyclic group of integer angles.

7. The Polynomial Ring and the Number-Theoretic Transform

7.1 From real attention to ring attention

Standard attention is a real-valued bilinear form on $\mathbb{R}^{d_k}$. We replace it with a bilinear form on the cyclotomic ring

$$R_q = \mathbb{Z}_q[x]/(x^N + 1), \quad N = 256.$$

For each Q, K vector we apply the CKKS-style encoder $e(v) = \lfloor \Delta \cdot v \rfloor$ with scaling factor Δ , lift to a polynomial in R_q , and compute the inner product as the coefficient of x^{N-1} in the (negacyclic) convolution $Q(x) \cdot K(\hat{x})$. Recovery is via $\langle q, k \rangle = \text{coeff}_{N-1}(\cdot)/\Delta^2$.

7.2 Theorem (KL-Zero Equivalence)

For $\Delta \geq 2^{10}$ and head dimension $d_k \leq N = 256$, the polynomial-ring attention is exact to floating-point ULP, and the KL divergence between the softmax distribution computed from real-valued logits and from ring-valued logits is zero.

Validated empirically on Gemma3 head dimension 256: $\text{KL}(\sigma_{\mathbb{R}} \parallel \sigma_{R_q}) = 0$, cosine = 1.

7.3 NTT for $O(N \log N)$ multiplication

The negacyclic NTT over a Proth prime $q \equiv 1 \pmod{2N}$ with primitive $2N$ -th root of unity ψ diagonalizes multiplication in R_q . We use the 60-bit prime $q = 576,460,752,312,401,921 = k \cdot 2^{16} + 1$ with $\psi = 1753$. Polynomial multiplication becomes

$$P \cdot Q = \text{NTT}^{-1}(\text{NTT}(P) \odot \text{NTT}(Q)).$$

7.4 Barrett reduction

Modular reduction mod q is replaced by the Barrett constant $\mu = \lfloor 2^{120}/q \rfloor$, eliminating division. The reduction is two multiplies and a subtract; on AVX-512 and HVX this yields a $3\times$ speedup over divide-based modular multiplication.

7.5 CRT-NTT: portability without `__int128`

Side B carried the CRT-NTT through to the engine: two 30-bit Proth primes q_1, q_2 run independent NTTs, and the result is recombined by Garner’s algorithm. Every intermediate is bounded above by $q_i^2 < 2^{60}$, which fits a `uint64`. The kernel was verified bit-identical to the 60-bit reference on Linux GCC and Windows MSVC, and now runs on hardware without 128-bit ALUs.

8. Poncelet Closure and the Layer Iteration

8.1 The Euler-Chapple relation

For two circles with radii R, r and center distance d , Poncelet’s closure theorem states that *if* a single triangle is inscribed in the outer and circumscribed about the inner,

then every triangle is. Euler’s relation $d^2 = R^2 - 2Rr$ characterises the closure condition. The deep generalisation: for any two conics in a plane, a polygon of n sides closes for one initial point iff it closes for every initial point, and the condition is $n\delta \equiv 0$ on an associated elliptic curve, where δ is the divisor class linking the two conics.

8.2 Hidden state as a point orbit

Treat the hidden state at layer l as a point $P_l \in E$. Each layer applies a small endomorphism, so

$$P_{l+1} = P_l + \delta_l, \quad \delta_l \in E(K).$$

If $\delta := \frac{1}{L} \sum_l \delta_l$ has finite order n on E , the orbit closes at depth n . Concretely:

- $n \mid L$: the residual stream completes integer cycles by the model’s full depth.
- $\text{ord}(\delta)$ small: an *early-exit* condition. Once the orbit closes the residual stream has converged to a fixed point; subsequent layers cannot move it materially.
- Length- p closed orbits in attention (with p prime) admit irreducible circulant representation, replacing $O(n^2)$ score computations by $O(n \log n)$.

8.3 Theorem (Caustic Invariance)

Let E act on the residual stream by point addition. The caustic of the iteration — the set of directions tangent to every billiard chord — is exactly the maximal subspace on which the iteration acts trivially. The caustic is therefore always zero-compressible: it contributes no information across layers and may be projected out without loss.

The caustic is the **null space of compression**: the directions that every layer leaves unchanged.

9. The Ulam-Sacks Spiral and Adaptive Precision

The Ulam spiral (integers laid out on a square spiral) and the Sacks spiral (integers laid out on a polar spiral with radius $\sqrt{n}$) both reveal that primes cluster along quadratic curves, most famously along $n^2 + n + 41$. In our setting:

- **V-cache adaptive precision.** Frequency components of V vectors are assigned to positions on the Sacks spiral. Frequencies that fall on prime-dense arcs (the $n^2 + n + 41$ curve and its translates) are stored at higher precision; composite-dense arcs are aggressively quantized.
- **Cold-start activation schedule.** The first 40 tokens of any sequence preferentially fire FFN neurons indexed by $n^2 + n + 41 \bmod d_{\text{ffn}}$. For the first 40 tokens these indices are guaranteed prime (Euler’s discovery), and therefore *primitive* in our UFD — the activation pattern is provably maximally diverse during cold start. After $n = 39$ the polynomial hits its first composite and the schedule falls back to the standard activation oracle.
- **Cramér gap prefetch.** Cramér’s conjecture predicts prime gaps near $\log^2 p$. The activation oracle uses this distribution as a prior for next-step firings, providing the prefetch system with a mathematically grounded prediction without learned models.

10. The Thirteen-Step Mapping

We summarise the complete mapping from token embedding through language-model head. Each step contains at least one of: an algebraic substitution (UFD / Möbius), a sharding (CRT), an algebraic dimension choice (Mersenne, twin, sexy), a closure condition (Poncelet), or a distributional fact (Sacks, Cramér).

#	Step	Algebraic Replacement
1	Embedding lookup	Möbius reconstruction over square-free token indices; CRT vocabulary sharding
2	RMSNorm (pre-attn)	Mersenne-prime scaling; Poncelet norm tracking $d^2 = R^2 - 2Rr$; early-exit on closure
3	Q/K/V projections	Twin-prime head pairing; sexy-prime GQA grouping; group-law inter-layer weight sharing
4	SP Write (KV $\rightarrow$ archive)	Poncelet closure as eviction trigger; CRT-sharded KV
5	FUSED_KQ	UFD-exact decompression; Heegner endomorphism commutes with group law
6	Softmax	p -adic exponential on integers; circulant attention on closed orbits
7	Fused V weighted sum	Spinor reconstruction across twin-paired heads; Sacks-spiral adaptive precision
8	Attention output projection	CRT decomposition of W_O into independent sub-matrices
9	FFN (ternary skeleton + sparse residual)	Mersenne-dimensional skeletons; twin-prime neuron pairs; n^2+n+41 cold-start schedule
10	Activation oracle update	Cramér prime-gap prefetch; Poncelet early exit
11	Residual add + norm	Group-law residual; UFD invertibility for RevNet-style reconstruction
12	Per-layer loop (master schedule)	$n\delta \equiv 0$ adaptive depth; caustic projection; periodic KV eviction
13	LM head	CRT pruning of vocabulary logits; Mersenne-prime sampling temperature; Sacks-spiral next-token prediction

11. The Grand Unified View

Stacking the thirteen steps reveals the architecture: **the transformer forward pass is a discrete dynamical system on an elliptic curve over a class-number-1 field.**

- The hidden state is a point on E/K .
- Each layer is a point addition.
- Each attention is a bilinear form on R_q .

- Each FFN is a structured sparse map indexed by primes of $\mathcal{O}_K$.
- The orbit closes at depth n iff $\text{ord}(\delta) \mid n$ — Poncelet.
- The caustic is the invariant subspace — always compressible.
- The Shannon limit of the architecture is the information content of the curve point: $\log_2 \text{ord}(\delta)$ bits, no more.

Everything else — every dimension, every head ratio, every quantization scheme — is determined by an algebraic structure of $\mathcal{O}_K$, not a hyperparameter sweep.

12. Verified Theorems and Extensions

The companion test suite (Part II §10) mechanically verifies the following at the time of submission:

- **T1 — Endomorphism realization.** The hidden state trajectory through L layers embeds in E^L exactly.
- **T2 — Möbius UFD compression.** Reconstruction over $\mathcal{O}_K$ at square-free basis is exact.
- **T3 — Hasse-Weil = Shannon limit.** $|\#E_p(\mathbb{F}_p) - (p + 1)| \leq 2\sqrt{p}$.
- **T4 — Frobenius cancellation.** §3.2 above; validated bit-identical at six significant figures on Gemma3-1B.
- **T5 — Deuring / CM Sato-Tate.** Asymmetric distribution of a_p between split and inert primes.
- **T6 — CRT exact sharding.** Two-prime kernel bit-identical to 60-bit reference; portable.
- **E9.1 — Stern-Brocot RoPE.** Discrepancy $\phi = 0.00134$ (CM endomorphism) $\ll 0.05576$ (standard RoPE).
- **E9.2 — Weil pairing on $E[n]$.** Miller’s algorithm validated, bilinearity confirmed.
- **E9.3 — Hecke multiplicativity.** 20/20 trials passed.
- **E9.5 — LLL reduction.** KV-write optimization, 20/20 trials.
- **E9.6 — BSD analytic rank.** L -function computation via Sage, all toy curves verified.
- **E10 — Iwasawa $\mu = 0$.** Linear $\text{ord-}p$ growth bound confirmed; residual stream depth-stable.

13. Conclusion

The choice of carrier matters. By replacing $\mathbb{R}^d$ with $\mathcal{O}_K$, and the implicit identity map of the residual stream with the group law on a CM elliptic curve, we obtain a transformer in which compression, normalization, sharding, activation, and decoding are no longer ad-hoc tricks — they are consequences of class number 1 and the splitting law in $\mathbb{Q}(\sqrt{-163})$. The dominant technical risk in the framework (that integer scaling would propagate through nonlinear steps and destroy the model) was eliminated by Theorem 3.2: Frobenius scale cancels through the RMSNorm boundary. The companion paper (Part II) reports the system that runs this mathematics on a phone.

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*This work was carried out over two days (2026-05-17 to 2026-05-19) on top of the Shannon-Prime project. The system that executes this mathematics is described in P